

CBEH Final Mock Examination

(Simulation 3)

Time Allowed: 90 Minutes

Total Questions: 70 (30 Cell Biology, 24 Histology, 12 Embryology, 4 Interdisciplinary)

Passing Requirement: Minimum 60% overall, with at least 50% in each module.

PART I: Cell Biology (30 Questions)

1. Multiple Choice: Which of the following transport mechanisms utilizes the concentration gradient of one molecule to drive the transport of another molecule against its gradient?

- A. Osmosis
- B. Primary active transport
- C. Simple diffusion
- D. Secondary active transport
- E. Facilitated diffusion via a channel

2. Open Question (Max 200 words): Endocytosis is a crucial cellular process for internalizing extracellular material. Briefly describe the process of receptor-mediated endocytosis, specifically mentioning the role of clathrin and endosomes.

3. True or False: Microfilaments (actin filaments) are primarily concentrated in the cell cortex just beneath the plasma membrane, where they determine the shape of the cell's surface and are essential for whole-cell locomotion.

4. Fill in the gap: Molecular _____ are proteins that assist in the proper folding of newly synthesized polypeptides, preventing inappropriate aggregation.

5. Matching: Match the type of intercellular signaling with its correct description.

- 1. Paracrine signaling
- 2. Endocrine signaling
- 3. Autocrine signaling
- 4. Contact-dependent signaling

A. A cell secretes a signal that binds to receptors on its own surface.

B. Hormones are secreted into the bloodstream and distributed widely throughout the

body.

C. A membrane-bound signal molecule on one cell binds to a receptor on an adjacent cell.

D. Local mediators are secreted into the extracellular fluid and act on nearby cells.

6. Multiple Choice: During DNA replication, the topological stress (supercoiling) generated ahead of the replication fork is relieved by which enzyme?

- A. DNA Helicase
- B. DNA Ligase
- C. Topoisomerase
- D. DNA Polymerase delta
- E. Primase

7. Multiple Choice: Telomerase is highly active in stem cells but inactive in most somatic cells. Its primary function is to:

- A. Splice introns out of pre-mRNA.
- B. Repair double-strand DNA breaks via homologous recombination.
- C. Add repetitive DNA sequences to the ends of linear chromosomes, solving the end-replication problem.
- D. Acetylate histones to promote chromatin decondensation.
- E. Excise mispaired nucleotides during DNA proofreading.

8. Open Question (Max 200 words): Discuss the differences in how transcription and translation are spatially and temporally coupled in prokaryotes versus eukaryotes.

9. Fill in the gap: During eukaryotic RNA processing, the addition of a _____ to the 3' end of the mRNA transcript protects it from enzymatic degradation and assists in export from the nucleus.

10. True or False: During translation, the incoming aminoacyl-tRNA first binds to the E (Exit) site of the ribosome before moving to the P (Peptidyl) site.

11. Multiple Choice: The tumor suppressor gene *p53* is frequently mutated in human cancers. A key physiological role of wild-type *p53* is to:

- A. Act as a transcription factor that upregulates the Cdk inhibitor p21 in response to DNA damage.
- B. Promote angiogenesis by upregulating VEGF.
- C. Act as a specific receptor tyrosine kinase for survival signals.
- D. directly degrade the Retinoblastoma (Rb) protein.
- E. Phosphorylate cell-cycle cyclins to advance the cell into S-phase.

12. Matching: Match the phase of mitosis with its defining cellular event.

- 1. Prophase

2. Prometaphase

3. Metaphase

4. Anaphase

A. Chromosomes align along the equator of the spindle.

B. Sister chromatids synchronously separate and are pulled toward opposite poles.

C. The nuclear envelope abruptly breaks down, allowing spindle microtubules to attach to kinetochores.

D. Replicated chromosomes condense, and the mitotic spindle begins to assemble outside the nucleus.

13. Multiple Choice: Which of the following mechanisms is responsible for the independent assortment of maternal and paternal homologs into gametes during meiosis?

A. The random alignment of bivalents at the metaphase plate during Meiosis I.

B. The action of the spliceosome during Prophase I.

C. The separation of sister chromatids during Anaphase II.

D. The breakdown of the nuclear envelope in Prometaphase I.

E. The specific segregation of the X and Y chromosomes.

14. Open Question (Max 200 words): In human genetics, phenotypes do not always perfectly correlate with genotypes. Explain the concepts of *penetrance* and *expressivity* as extensions to Mendel's laws.

15. Fill in the gap: The organized visual profile of an individual's complete set of metaphase chromosomes, arranged by size and shape to diagnose numerical or large structural anomalies, is called a _____.

16. Multiple Choice: A point mutation that results in the substitution of one amino acid for another in the final protein sequence is known as a:

A. Silent mutation

B. Nonsense mutation

C. Missense mutation

D. Frameshift mutation

E. Promoter mutation

17. True or False: Proto-oncogenes generally require a dominant, "gain-of-function" mutation in only one allele to promote oncogenesis, whereas tumor suppressor genes typically require recessive, "loss-of-function" mutations in both alleles.

18. Multiple Choice: Apoptosis is executed by a cascade of specialized intracellular proteases called:

A. Caspases

B. Polymerases

- C. Cyclins
- D. Kinases
- E. Hydrolases

19. Matching: Match the viral characteristic/cycle with the correct description.

- 1. Enveloped virus
- 2. Naked virus
- 3. Retrovirus
- 4. Viral tropism

- A. A virus lacking a lipid bilayer, consisting only of a nucleocapsid.*
- B. The specificity of a virus for a particular host tissue or cell type.*
- C. An RNA virus that uses reverse transcriptase to integrate into the host genome.*
- D. A virus possessing a lipid bilayer derived from the host cell membrane.*

20. Open Question (Max 200 words): Describe the differences between embryonic stem cells (ESCs) and induced pluripotent stem cells (iPSCs) regarding their cellular source and potency. How can iPSCs be utilized for disease modeling?

21. Fill in the gap: The CRISPR-Cas9 system utilizes a specific _____ RNA to direct the Cas9 endonuclease to the precise genomic sequence designated for editing.

22. Multiple Choice: Proteins that are destined to reside in the lumen of the Endoplasmic Reticulum (ER) possess a specific C-terminal retention signal sequence known as:

- A. The NLS (Nuclear Localization Signal)
- B. The KDEL sequence
- C. Mannose-6-phosphate
- D. The poly-A tail
- E. Ubiquitin

23. Multiple Choice: The shape and structural stability of the nuclear envelope are provided by a meshwork of intermediate filaments located on its inner face, called the:

- A. Nuclear lamina
- B. Centrosome
- C. Nucleolus
- D. Kinetochore
- E. Actin cortex

24. True or False: The Smooth Endoplasmic Reticulum (SER) is highly developed in cells specialized for the synthesis of steroid hormones and the detoxification of lipid-soluble drugs.

25. Multiple Choice: Lysosomal storage diseases (e.g., Tay-Sachs disease) are primarily caused by:

- A. The inability of the cell to form autophagosomes.
- B. Mutations that over-activate the proteasome.
- C. A genetic deficiency in one or more acid hydrolase enzymes within the lysosomes.
- D. The failure to append the KDEL sequence to lysosomal enzymes.
- E. Defective mitochondrial electron transport chains.

26. Matching: Match the level of protein structure with its definition.

- 1. Primary structure
- 2. Secondary structure
- 3. Tertiary structure
- 4. Quaternary structure

- A. The overall 3D conformation of a single polypeptide chain.*
- B. The linear sequence of amino acids linked by peptide bonds.*
- C. The assembly of multiple distinct polypeptide subunits into a functional protein complex.*
- D. Local folding patterns, such as alpha-helices and beta-sheets, stabilized by hydrogen bonds.*

27. Open Question (Max 200 words): Explain how Cyclin-Dependent Kinases (CDKs) are regulated during the cell cycle. Mention the specific role of cyclins and the degradation mechanisms that ensure the cell cycle progresses in one direction.

28. Fill in the gap: The specific DNA sequence where RNA polymerase binds to initiate transcription is called the _____.

29. Multiple Choice: In the fluid-mosaic model, integral membrane proteins:

- A. Float freely in the cytoplasm until needed.
- B. Are weakly attached to the surface of the lipid bilayer via non-covalent interactions.
- C. Traverse the lipid bilayer and typically contain hydrophobic alpha-helices.
- D. Form the lipid rafts exclusively.
- E. Are synthesized entirely inside the nucleus.

30. Multiple Choice: The failure of homologous chromosomes or sister chromatids to separate properly during meiosis is known as:

- A. Crossing-over
 - B. Independent assortment
 - C. Recombination
 - D. Nondisjunction
 - E. Translocation
-

PART II: Histology (24 Questions)

31. Multiple Choice: The most commonly used histological staining technique, which stains acidic structures (like DNA) blue/purple and basic structures (like cytoplasm) pink/red, is:

- A. Periodic Acid-Schiff (PAS)
- B. Silver stain
- C. Hematoxylin and Eosin (H&E)
- D. Masson's trichrome
- E. Osmium tetroxide

32. True or False: Hemidesmosomes are anchoring junctions that attach the basal domain of epithelial cells to the underlying basement membrane using integrins.

33. Fill in the gap: Exocrine glands that discharge their secretory vesicles by “pinching off” the apical portion of the cell (e.g., mammary glands) utilize an _____ mechanism of secretion.

34. Open Question (Max 200 words): Contrast the histological structure and mechanical properties of loose (areolar) connective tissue with dense regular connective tissue. Provide one anatomical location for each.

35. Multiple Choice: Which type of adipose tissue is specialized for non-shivering thermogenesis (heat production) and is histologically multilocular?

- A. Yellow adipose tissue
- B. White adipose tissue
- C. Fibrous adipose tissue
- D. Brown adipose tissue
- E. Reticular adipose tissue

36. Matching: Match the type of cartilage with its primary feature/location.

1. Hyaline cartilage
2. Elastic cartilage
3. Fibrocartilage

A. Characterized by abundant type I collagen; found in intervertebral discs and the pubic symphysis.

B. Contains a dense network of elastin fibers; found in the epiglottis and pinna of the ear.

C. The most common type, providing low-friction surfaces in synovial joints and forming the fetal skeleton.

37. Multiple Choice: In mature compact bone, the concentric lamellae are organized around a central canal containing blood vessels and nerves. This functional unit is called a(n):

- A. Lacuna
- B. Osteon (Haversian system)
- C. Trabecula
- D. Canaliculus
- E. Volkmann's canal

38. Fill in the gap: Platelets are anucleate cytoplasmic fragments derived from giant polyploid cells in the bone marrow known as _____.

39. True or False: The thymus is a primary lymphoid organ where B-lymphocytes undergo maturation and selection.

40. Open Question (Max 200 words): Describe the sliding filament mechanism of skeletal muscle contraction. Mention the roles of the sarcomere, actin, myosin, and calcium ions.

41. Multiple Choice: In smooth muscle cells, actin filaments anchor to structure equivalents of the skeletal Z-discs, which are called:

- A. Intercalated discs
- B. Sarcoplasmic reticulums
- C. Dense bodies
- D. Caveolae
- E. Motor endplates

42. Matching: Match the neuroglial cell with its function.

- 1. Astrocyte
- 2. Microglia
- 3. Ependymal cell
- 4. Schwann cell

A. Myelinates single axons in the Peripheral Nervous System.

B. Lines brain ventricles and produces cerebrospinal fluid.

C. Acts as the resident macrophage of the Central Nervous System.

D. Maintains the blood-brain barrier and regulates the chemical environment of neurons.

43. Multiple Choice: In myelinated nerve fibers, the bare gaps between adjacent myelin sheaths, which are enriched in voltage-gated sodium channels for saltatory conduction, are:

- A. Synaptic clefts

- B. Schmidt-Lanterman clefts
- C. Neuromuscular junctions
- D. Nodes of Ranvier
- E. Axon hillocks

44. Fill in the gap: _____ glands lack a duct system and secrete their products (hormones) directly into the surrounding interstitial fluid and bloodstream.

45. True or False: Pseudostratified ciliated columnar epithelium actually consists of a single layer of cells, all of which contact the basement membrane, though not all reach the apical surface.

46. Multiple Choice: Which microscopy technique enhances the contrast of transparent, unstained living cells by exploiting differences in their refractive indices?

- A. Phase-contrast microscopy
- B. Scanning electron microscopy
- C. Fluorescence microscopy
- D. Transmission electron microscopy
- E. Bright-field microscopy

47. Multiple Choice: Which leukocyte is characterized by a bilobed nucleus and large specific granules that stain brightly red/orange with eosin, playing a major role in parasitic infections?

- A. Neutrophil
- B. Basophil
- C. Monocyte
- D. Lymphocyte
- E. Eosinophil

48. Matching: Match the connective tissue cell type with its function.

- 1. Fibroblast
- 2. Macrophage
- 3. Mast cell
- 4. Plasma cell

- A. Secretes large amounts of antibodies (immunoglobulins).*
- B. Secretes histamine and heparin during inflammatory responses.*
- C. Phagocytizes debris and presents antigens.*
- D. Synthesizes collagen, elastin, and ground substance.*

49. Open Question (Max 200 words): Describe the histological and functional differences between the red pulp and the white pulp of the spleen.

50. Fill in the gap: The large, multinucleated cell responsible for the resorption and breakdown of bone matrix during bone remodeling is the _____.

51. Multiple Choice: Cartilage can grow by two mechanisms. The process by which new cartilage is deposited on the surface of existing cartilage by chondroblasts in the perichondrium is called:

- A. Interstitial growth
- B. Endochondral ossification
- C. Appositional growth
- D. Intramembranous ossification
- E. Hypertrophy

52. True or False: At a chemical synapse, the transmission of a nerve impulse is bidirectional and relies on gap junctions to pass ions directly between neurons.

53. Multiple Choice: Which type of cell junction allows for direct electrical and metabolic communication by permitting the passage of small ions and molecules between adjacent cells?

- A. Desmosome
- B. Tight junction (Zonula occludens)
- C. Gap junction
- D. Adherens junction
- E. Hemidesmosome

54. Multiple Choice: Histologically, which feature is unique to cardiac muscle tissue and is not found in skeletal or smooth muscle?

- A. Central nuclei
 - B. Intercalated discs
 - C. Actin and myosin filaments
 - D. Spindle-shaped cells
 - E. Peripheral nuclei
-

PART III: Embryology (12 Questions)

55. Multiple Choice: In the female reproductive cycle, the surge of Luteinizing Hormone (LH) directly triggers which event?

- A. Menstruation
- B. Degeneration of the corpus luteum
- C. Ovulation
- D. Implantation
- E. Formation of the primary follicle

56. Fill in the gap: The terminal phase of male gametogenesis, where round spermatids undergo dramatic morphological changes (including acrosome formation and flagellum development) to become mature spermatozoa, is known as _____.

57. True or False: The “zona reaction” occurs immediately after the first sperm penetrates the oocyte, rendering the zona pellucida impermeable to other sperm to prevent polyspermy.

58. Matching: Match the primary germ layer with one of its major developmental derivatives.

1. Ectoderm
2. Mesoderm
3. Endoderm

A. Forms the epithelial lining of the gastrointestinal and respiratory tracts.

B. Gives rise to the cardiovascular system, muscles, and bones.

C. Gives rise to the epidermis of the skin and the entire nervous system.

59. Open Question (Max 200 words): During the 4th week of human development, the embryo undergoes significant folding. Explain the outcome of lateral folding and craniocaudal folding, specifically referring to the formation of the body wall and the primitive gut tube.

60. Multiple Choice: The fetal portion of the placenta, which features chorionic villi that interface with maternal blood, develops primarily from the:

- A. Hypoblast
- B. Inner cell mass (embryoblast)
- C. Trophoblast
- D. Epiblast
- E. Amnion

61. Multiple Choice: Congenital malformations such as omphalocele occur due to a failure in the normal developmental sequence of the primitive gut. Normal development of the midgut requires it to:

- A. Herniate into the umbilical cord and undergo a 270-degree rotation before returning to the abdominal cavity.
- B. Fuse completely with the neural tube.
- C. Remain entirely within the pelvic cavity throughout gestation.
- D. Degenerate and be replaced by the urogenital sinus.
- E. Differentiate directly into the diaphragm.

62. Fill in the gap: The _____ _____ cells are a highly migratory population that detach from the lateral borders of the neural folds during neurulation, giving rise to

peripheral ganglia, melanocytes, and facial mesenchyme.

63. True or False: The blastocyst consists of an outer cell mass that will form the embryo proper, and an inner cell mass that will form the placenta.

64. Open Question (Max 200 words): Compare the timing and continuity of meiosis in human males versus females. Mention when meiosis initiates, the occurrence of arrest phases, and when the process is completed in each sex.

65. Multiple Choice: During the 4th week of development, the paraxial mesoderm segments into block-like structures alongside the neural tube. These structures, which will give rise to the axial skeleton, skeletal muscle, and dermis, are called:

- A. Pharyngeal arches
- B. Somites
- C. Placodes
- D. Neuropores
- E. Villi

66. Multiple Choice: Myelination of peripheral nerves is critical for proper nervous system maturation. Which embryonic cell lineage gives rise to the Schwann cells responsible for this process?

- A. Endoderm
 - B. Neural tube ectoderm
 - C. Neural crest cells
 - D. Lateral plate mesoderm
 - E. Notochord
-

PART IV: Interdisciplinary (4 Questions)

67. Multiple Choice: *[Embryology + Histology]* Developmental defects such as *situs inversus* and Kartagener syndrome are caused by improper fluid flow during embryogenesis. This pathology is directly linked to a histological and cellular defect in which of the following?

- A. Gap junction connexins
- B. Motile cilia (dynein arm defects in microtubules)
- C. Collagen type II synthesis in cartilage
- D. Myosin heavy chains in cardiac muscle
- E. Voltage-gated sodium channels in neurons

68. Open Question (Max 200 words): *[Stem Cells + Embryology]* Induced pluripotent stem cells (iPSCs) are utilized for disease modeling. If a researcher is modeling a neural tube

closure defect (e.g., spina bifida), explain what embryonic cell lineage the iPSCs must be guided to differentiate into. Furthermore, describe the basic morphogenetic movements those cells normally undergo during the 3rd and 4th weeks of development (neurulation).

69. Open Question (Max 200 words): *[Cell Biology + Histology + Embryology]*

Epidermolysis bullosa (EB) is a disease characterized by extreme skin fragility. From a cell biology perspective, which cytoskeletal filament is mutated in *Epidermolysis bullosa simplex*? Histologically, which cell junction fails as a result, and how does this affect the attachment of the epidermis to the underlying connective tissue?

70. Matching: *[Apoptosis + Histology/Embryology]* Match the physiological or pathological event with its corresponding role of apoptosis.

1. Separation of webbed fingers and toes during embryonic development.
2. Sloughing off of the inner uterine lining during menstruation.
3. Elimination of T-lymphocytes that strongly bind to self-antigens in the thymus.
4. Death of neurons due to severe ischemic injury (stroke) resulting in membrane rupture and inflammation.

- A. Not apoptosis; this describes necrosis.*
B. Physiologic apoptosis for immune tolerance/selection.
C. Physiologic apoptosis for morphogenetic sculpting.
D. Physiologic apoptosis driven by hormonal withdrawal.
-

ANSWER KEY

Part I: Cell Biology

1. **D**
2. **Open Concept:** Receptor-mediated endocytosis begins when extracellular ligands bind to specific transmembrane receptors. These receptor-ligand complexes cluster in specific membrane regions coated on the cytosolic side by the protein clathrin. The clathrin lattice invaginates and pinches off to form a coated vesicle. The vesicle loses its clathrin coat and fuses with an early endosome, where the acidic environment often causes the receptor to release its cargo.
3. **True**
4. **Chaperones**

5. **Matching:** 1-D, 2-B, 3-A, 4-C
6. **C**
7. **C**
8. **Open Concept:** In prokaryotes, transcription and translation occur in the same compartment (the cytoplasm) simultaneously; ribosomes begin translating the mRNA while RNA polymerase is still transcribing it. In eukaryotes, these processes are spatially and temporally separated: transcription occurs in the nucleus, the pre-mRNA undergoes extensive processing (splicing, capping, polyadenylation), and the mature mRNA is exported to the cytoplasm where translation occurs.
9. **Poly-A tail** (or polyadenylate tail)
10. **False** (The incoming aminoacyl-tRNA binds to the A (Aminoacyl) site, not the E site).
11. **A**
12. **Matching:** 1-D, 2-C, 3-A, 4-B
13. **A**
14. **Open Concept:** Penetrance refers to the proportion of individuals carrying a specific allele that actually express the associated phenotype (e.g., incomplete penetrance means some mutation carriers show no symptoms). Expressivity refers to the degree or severity to which a phenotype is expressed in individuals who have the trait (e.g., varying severity of symptoms among patients with the same mutation).
15. **Karyotype**
16. **C**
17. **True**
18. **A**
19. **Matching:** 1-D, 2-A, 3-C, 4-B
20. **Open Concept:** ESCs are derived from the inner cell mass of the blastocyst and are naturally pluripotent. iPSCs are generated by taking adult somatic cells (like fibroblasts) and reprogramming them back to a pluripotent state using specific transcription factors. iPSCs can be used for disease modeling by taking cells from a patient with a genetic disorder, reprogramming them into iPSCs, and then differentiating them in vitro into the affected cell type (e.g., neurons) to study the pathology in a dish.

21. **Guide** (or gRNA)
22. **B**
23. **A**
24. **True**
25. **C**
26. **Matching:** 1-B, 2-D, 3-A, 4-C
27. **Open Concept:** CDKs are continuously present in the cell but are only active when bound to their specific regulatory proteins, cyclins. Cyclin levels rise and fall cyclically. Progression through checkpoints is driven by Cyclin-CDK complex activation. To ensure the cell cycle is irreversible, cyclins are targeted for destruction by the ubiquitin-proteasome system (e.g., via the APC/C complex), abruptly shutting off CDK activity and pushing the cell into the next phase.
28. **Promoter**
29. **C**
30. **D**

Part II: Histology

31. **C**
32. **True**
33. **Apocrine**
34. **Open Concept:** Loose (areolar) connective tissue has a high ratio of ground substance and cells to fibers, making it flexible, highly vascularized, and ideal for diffusion and immune defense (e.g., lamina propria beneath epithelia). Dense regular connective tissue consists mostly of tightly packed, parallel bundles of type I collagen with very little ground substance and few fibroblasts, providing immense tensile strength in one direction (e.g., tendons and ligaments).
35. **D**
36. **Matching:** 1-C, 2-B, 3-A
37. **B**
38. **Megakaryocytes**

39. **False** (The thymus is for T-lymphocyte maturation; B-lymphocytes mature in the Bone marrow).
40. **Open Concept:** A sarcomere is the contractile unit of skeletal muscle. When a muscle is stimulated, calcium ions are released from the sarcoplasmic reticulum and bind to troponin, exposing myosin-binding sites on the thin actin filaments. Myosin heads (thick filaments) bind to actin and use ATP hydrolysis to pivot (the power stroke), pulling the actin filaments toward the center of the sarcomere, shortening the muscle fiber without the filaments themselves changing length.
41. **C**
42. **Matching:** 1-D, 2-C, 3-B, 4-A
43. **D**
44. **Endocrine**
45. **True**
46. **A**
47. **E**
48. **Matching:** 1-D, 2-C, 3-B, 4-A
49. **Open Concept:** In the spleen, the white pulp consists of lymphoid nodules and periarteriolar lymphoid sheaths (PALS) surrounding central arteries; its function is immunologic, initiating responses to blood-borne antigens. The red pulp consists of splenic cords (of Billroth) and venous sinusoids; its histological structure allows macrophages to act as a filter to detect, phagocytize, and destroy old or damaged erythrocytes.
50. **Osteoclast**
51. **C**
52. **False** (Chemical synapses are unidirectional and use neurotransmitters. Gap junctions are found in electrical synapses).
53. **C**
54. **B**

Part III: Embryology

55. **C**

56. **Spermiogenesis**

57. **True**

58. **Matching:** 1-C, 2-B, 3-A

59. **Open Concept:** Lateral folding brings the edges of the flat embryonic disc together ventrally to form a cylindrical body wall, enclosing the intraembryonic coelom. Craniocaudal (head and tail) folding curves the embryo into a fetal position. Together, these folding movements incorporate the dorsal portion of the yolk sac into the embryo, forming the primitive gut tube (foregut, midgut, hindgut) lined by endoderm.

60. **C**

61. **A**

62. **Neural crest**

63. **False** (The outer cell mass/trophoblast forms the placenta, while the inner cell mass/embryoblast forms the embryo proper).

64. **Open Concept:** In females, meiosis initiates during fetal development, but primary oocytes arrest in Prophase I (diplotene) until puberty. Following puberty, a few oocytes resume meiosis each month but arrest again at Metaphase II; meiosis is only completed if fertilization occurs. In males, meiosis begins at puberty and is a continuous, uninterrupted process from spermatocyte to spermatid, continuing throughout the male's lifetime.

65. **B**

66. **C**

Part IV: Interdisciplinary

67. **B**

68. **Open Concept:** To model neural tube defects, iPSCs must be directed to differentiate into embryonic **ectoderm**, specifically the **neuroectoderm**. During normal neurulation (weeks 3-4), the notochord induces the overlying ectoderm to thicken into the neural plate. The lateral edges of the plate elevate to form neural folds, which then converge and fuse in the dorsal midline to form the neural tube. Failure of this fusion results in defects like spina bifida.

69. **Open Concept:** In Epidermolysis bullosa simplex, there are mutations in the genes encoding **keratin** intermediate filaments. Histologically, keratin filaments are anchored to the basal lamina via **hemidesmosomes**. When keratin is defective, the mechanical integrity of the basal epithelial cells collapses. Upon mild friction, the hemidesmosomal attachments fail or the basal cells tear, causing the epidermis to separate from the underlying connective tissue, forming painful blisters.

70. **Matching:** 1-C, 2-D, 3-B, 4-A